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A simulation of the quasi-standing wave and generalized half-wave loss of electromagnetic wave in non-ideal media

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Abstract

In this paper, the phenomena of reflection towards normal incident electromagnetic wave on double loss-medium are analyzed, and the superposition wave mechanism that is similar to standing wave in transparent medium is exposed. In this case, it is more suitable that wave impedance of media is used to determine the thin or thick medium and half-wave loss. Based on the thought mentioned above, the new concept of quasi-standing wave and generalized half-wave loss are put forward for the first time. Moreover, the basic criteria of generalized half-wave loss are given. Especially, the basic characters and generalized law of quasi-standing wave is simulated by means of computations, and their physical mechanisms are also described.
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Keywords: Electromagnetic wave; Quasi-standing wave; Generalized half-wave loss; Wave impedance, Materials design

1. Introduction

In some special application fields, it is important to suppress microwave reflection from metal structure targets [1,2]. For example, electromagnetic interference is drawing forth more attention, due to the explosive growth in the utilization of electrical and electronic devices in industrial, commercial, and military applications [3–5]. In fact, the main problem for the design of a magnetic absorber is related to the choice of the materials [6,7]. Meanwhile, electromagnetic property depends on magnetic or dielectric parameters such as permeability, permittivity and loss constants [8–11]. Actually, the search for design curves and equations for radar absorbing materials has been the subject of theoretical efforts for a long time.

According to the theory of classical electromagnetic field, when wave propagates in media from thin medium to thick medium, the reflection will take place at the interface between media, where the phase of reflection wave changes abruptly at half-period. This is the so-called half-wave loss. At this time, superposition of incident and reflection wave will form standing wave due to interference. Because of half-wave loss, a standing-wave node will appear at the interface. On the contrary, when wave propagates in media from thick medium to thin one, the standing wave will also be formed in the thick medium. Meanwhile, a standing-wave antinode will stand on the interface for which there is no half-wave loss. For transparent medium or little loss medium, the half-wave loss and standing wave have already been revealed [12–14].

In fact, with the propagation of electromagnetic wave, there exists some dielectric loss or magnetic loss in media due to various complex mechanisms, by which the media act on wave. Meanwhile, the ideal standing-wave will not exist in media. Moreover, so-called thin and thick medium can hardly be characterized by refractive index. Therefore, the concept of half-wave loss needs defining again. In this paper, the concept of generalized half-wave loss is introduced by means of wave impedance. Of course, the basic criteria are also provided. Furthermore, the superposition effect of reflection and incident wave nearby interface of loss media is studied in detail. It has been shown that the superposition wave is similar to standing wave. However, there are

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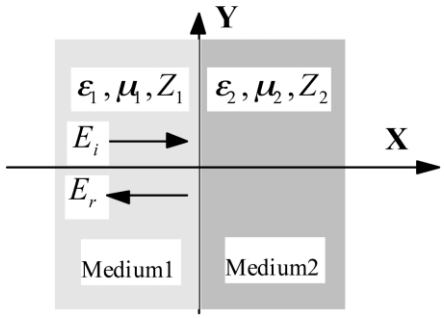


Fig. 1. The diagram of reflection at the interface of two-media.

some obvious differences towards the wave. Thus, authors define it as quasi-standing wave. In addition, the basic characteristics of quasi-standing wave are analyzed.

2. The reflection of electromagnetic wave at the interface of loss media

For the case of loss media, suppose that there exists normal incident electromagnetic wave. *Y*-axis is the interface of two homogeneous semi-infinite media. See Fig. 1.

Then, the electromagnetic parameters of media are $(\varepsilon_1,\mu_1,Z_1)$ and $(\varepsilon_2,\mu_2,Z_2)$ separately. Here, the planar wave of vertical polarization may be written by the following expression.

$$E_i = E_0 e^{-\gamma x},\ H_i = \frac{E_0}{Z_1} e^{-\gamma x} \tag{1}$$

Here, $\gamma = \alpha + i\beta$ stands for electromagnetic propagation constant, while α is attenuation factor of media, and β is phase factor of media. Their analytical expressions are as follows [5]

$$\alpha = \frac{\omega}{c}\sqrt{\mu'\varepsilon'}\sqrt{\frac{1}{2}[tg\delta_e tg\delta_m - 1 + \sqrt{(1+tg^2\delta_e)(1+tg^2\delta_m)}]} \tag{2}$$

$$\beta = \frac{\omega}{c}\sqrt{\mu'\varepsilon'}\sqrt{\frac{1}{2}[1 - tg\delta_e tg\delta_m + \sqrt{(1+tg^2\delta_e)(1+tg^2\delta_m)}]} \tag{3}$$

Where, $tg\delta_e = \varepsilon''/\varepsilon'$, $tg\delta_m = \mu''/\mu'$, respectively, stands for the tangent of dielectric loss angle and magnetic loss angle. Suppose *R* is reflection coefficient, then, the reflection wave at the interface of media may be written as follows:

$$E_r = E_0 \mathrm{Re}^{\gamma x},\ H_r = \frac{E_0}{Z_1}\mathrm{Re}^{\gamma x} \tag{4}$$

Here, the formula of reflection coefficient is

$$R = R' - iR'' = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{Z_2' - Z_1' + Z_2''^2 - Z_1''^2}{(Z_2' + Z_1')^2 + (Z_2'' + Z_1'')^2} - i\frac{2(Z_1'Z_2'' - Z_1''Z_2')}{(Z_2' + Z_1')^2 + (Z_2'' + Z_1'')^2} \tag{5}$$

According to Euler's formula, the formula above can also be written as follow

$$R = |R|e^{i\varphi} \tag{6}$$

Where, $|R|$ and $tg\varphi$ are represented by the following expressions

$$|R| = \sqrt{R'^2 + R''^2} \tag{7}$$

$$tg\varphi = \frac{R''}{R'} = \frac{2(Z_1'Z_2'' - Z_1''Z_2')}{Z_2' - Z_1' + Z_2''^2 - Z_1''^2} \tag{8}$$

Here, Z' and Z'' stand for real and imaginary parts of intrinsic wave-impedance. They are

$$Z' = Z_0\sqrt{\mu'/\varepsilon'}\frac{1}{\sqrt{1+tg^2\delta_e}} \times \sqrt{\frac{1}{2}[1 + tg\delta_e tg\delta_m + \sqrt{(1+tg^2\delta_e)(1+tg^2\delta_m)}]} \tag{9}$$

$$Z'' = Z_0\sqrt{\mu'/\varepsilon'}\frac{1}{\sqrt{1+tg^2\delta_e}} \times \sqrt{\frac{1}{2}[-(1 + tg\delta_e tg\delta_m) + \sqrt{(1+tg^2\delta_e)(1+tg^2\delta_m)}]} \tag{10}$$

3. Model and simulating of quasi-standing wave and generalized half-wave in loss media

3.1. Model of quasi-standing wave in loss media

Consider the composition of incident wave and reflection wave in Fig. 1, the total field will become

$$E = E_i + E_r = (\mathrm{e}^{-\mathrm{y}x} + \mathrm{Re}^{yx}) \tag{11}$$

It can be written as complex number expression

$$E = E_0[e^{-\alpha x}\cos\beta x + |R|e^{\alpha x}\cos(\beta x + \varphi)] + iE_0[|R|e^{-\alpha x}\sin(\beta x + \varphi) - e^{\alpha x}\sin\beta x] \tag{12}$$

According to Euler formula, the total field can be written as follows:

$$E = |E|e^{i\theta} \tag{13}$$

Here, $|E|$ and $tg\theta$ are

$$|E| = E_0\sqrt{e^{-2\alpha x} + |R|^2 e^{2\alpha x} + 2|R|\cos(2\beta + \varphi)} \tag{14}$$

$$tg\theta = \frac{|R|e^{-\alpha x}\sin(\beta x + \varphi) - e^{\alpha x}\sin\beta x}{|R|e^{\alpha x}\cos(\beta x + \varphi) + e^{-\alpha x}\cos\beta x} \tag{15}$$

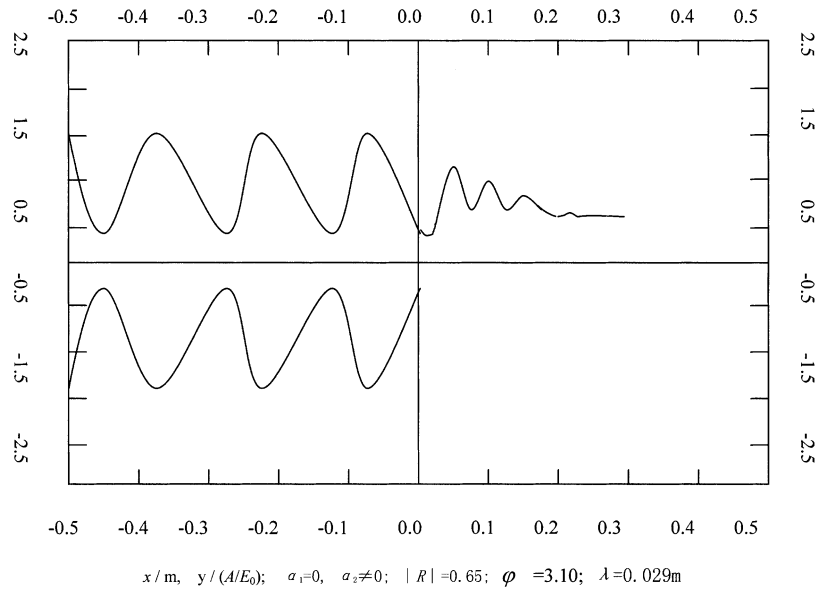


Fig. 2. Computer simulating towards quasi-standing wave in double media.

Formula (13) shows the composition of incident wave and reflection wave in medium 1. Obviously, relative amplitude $A=E/E_0$ depends on α, β, φ, x and R. In other words, it is the function of those parameters. Based on the facts above, authors define the standing wave as quasi-standing wave.

3.2. Simulating of quasi-standing wave in loss media

Figs. 2–5 show the branch of superposition wave curve simulated by computer. It has shown that the superposition is similar to ideal standing wave. There are several obvious features in these figures. The typical characters are as follows

1. The maximum values of superposition correspond to antinodes; while the minimum values of superposition correspond to wave nodes. The distance between nearby antinodes is half of wavelength.
2. Because of the influence of medium loss and reflection wave, the amplitude of superposition at wave node is not zero value. In other words, the mass

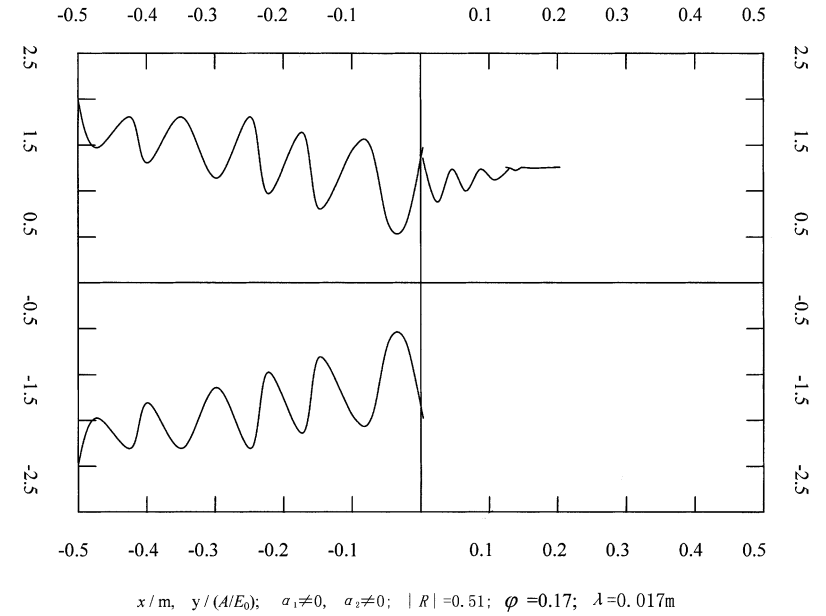


Fig. 3. Computer simulating towards quasi-standing wave in double media.

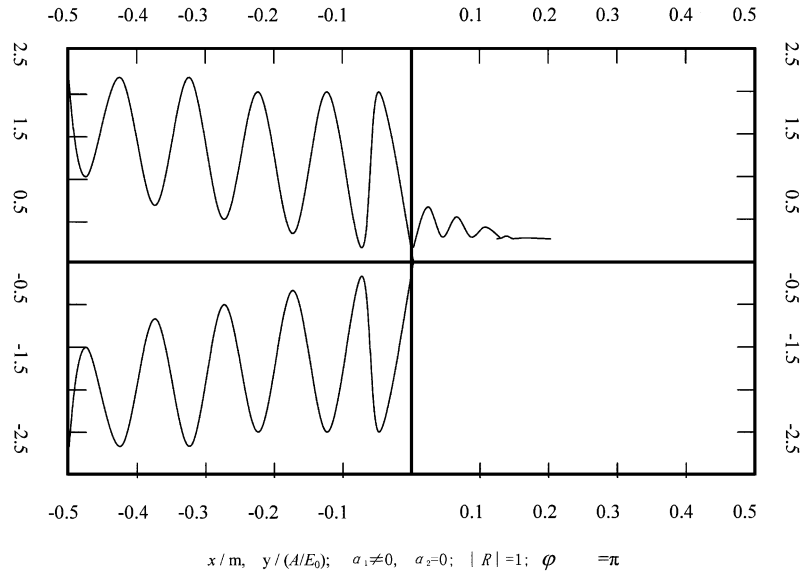


Fig. 4. Computer simulating towards quasi-standing wave in double media.

points at wavenode tend to vibrate.

3. Along the inverse direction of x-axis, the amplitudes of wave nodes and antinodes will decrease as the value of $|x|$ increases. The curves tend to be inclining towards upper-left. The reason for these phenomena probably is that the loss effects are different on incidence and reflection wave.
4. The difference between wave node and antinodes tends to decrease as the value decreases. It means that the standing wave exists in near field at interface, while the far field tends to be disappearing.

3.3. The criteria of generalized half-wave loss at loss medium interface

According to classical electromagnetic theory, when there exists half-wave loss, jumping phase by π will take place. In other words, one wave node will form at the interface.

$$A = \sqrt{1 + |R|^2 + 2|R|\cos\varphi} \tag{16}$$

Obviously, only if the phase difference equals π, the

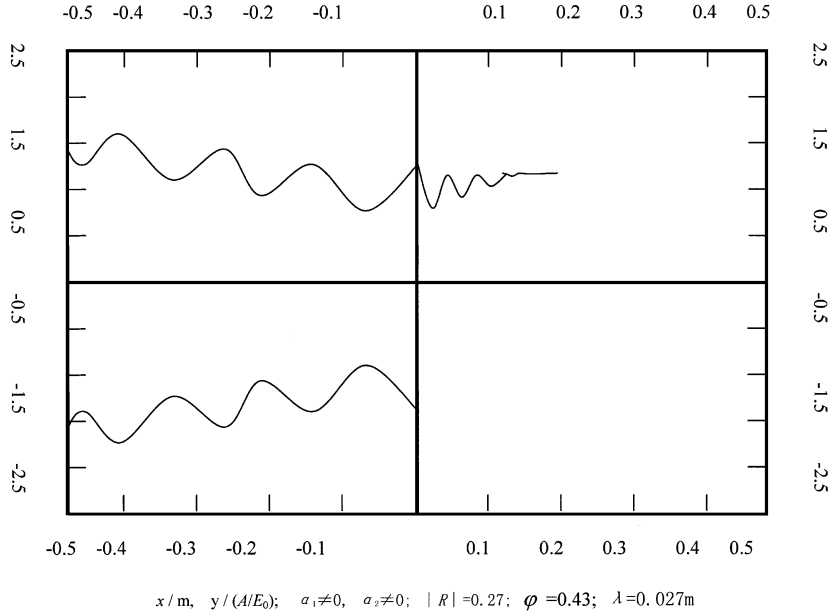


Fig. 5. Computer simulating towards quasi-standing wave in double media.

amplitude of superposition reaches minimum value. That is,

$$A = 1 - |R| \tag{17}$$

At this time, the follow conditions are required according to formula (6)

$$tg\pi = \frac{2(Z_1' Z_2'' - Z_1'' Z_2')}{Z_2' - Z_1' + Z_2''^2 - Z_1''^2} \tag{18}$$

The formula above can also be written

$$Z_1' Z_2'' - Z_1'' Z_2' = 0 \tag{19}$$

$$|Z_1| < |Z_2| \tag{20}$$

Formula (19) and (20) are just as half-wave loss criteria of electromagnetic wave in loss media. Based on the discussion above, the generalized half-wave loss can be introduced. It may be defined as follows. When electromagnetic wave propagates in medium from small $|Z|$ to large one, reflection probably shows half-wave loss. However, only if formula (18) exists, the effect takes place. Actually, half-wave loss in transparent medium can also be judged by $|Z_1| < |Z_2|$. In other words, the latter is only a special example of generalized half-wave loss in loss medium. Especially, the antinodes and wave nodes can be determined by the following formulas:

$$2\beta x + \varphi = (2n+1)\pi \ \textit{antinodes} \tag{21}$$

$$2\beta x + \varphi = 2n\pi \ \textit{wave nodes} \tag{22}$$

4. Conclusion

The simulation results have shown that the superposition of incident wave and reflection wave has obvious characteristics of quasi-standing wave. However, with the change of phase angle, the position of standing-wave node will be rather different. Moreover, it is not suitable that refraction index is used to judge half-wave loss. Inversely, wave impedance should be introduced. This is a criterion of generalized half-wave loss. The important things are that generalized half-wave loss and quasi-standing wave are strong theoretical foundations to reveal mechanism of multiplayer RAM. It is helpful for us to design absorber of lightweight and broadband against microwave.

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